

# Understanding Chess Structures

Using Unsupervised Learning

Clustering Early-Game Patterns from Lichess Games

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MTH 4224 – Intro to Machine Learning

Spring 2026

### Main idea

We are not predicting winner. We are studying structure in chess games.

- Chess games do not have true style labels.
- We use unsupervised learning to see if natural groups appear.
- Features describe early-game activity, structure, opening family, and volatility.
- Main question is whether games form clean groups or continuous tendencies.



### Main ML Question

Do chess games naturally cluster into tactical, positional, or balanced groups?

*The goal is exploratory structure discovery, not supervised prediction.*

- ① Do chess games form natural clusters when described by engineered early-game features?
- ② Are clusters more influenced by dynamic features or structural features?
- ③ Can DBSCAN identify sparse or unusual regions better than general clustering methods?

### Important point

Since there are no true style labels, silhouette score and interpretability must be used together.

- Dataset: **10,000 Lichess PGN games**
- Final data after filtering: **8,621 games**
- First **20 full moves** used
- Unsupervised learning: no outcome labels
- Final clustering matrix: **43 numerical features**

### Feature types

- Dynamic: captures, checks, volatility
- Structural: pawns, center control, king danger
- Opening: ECO family one-hot features

**10,000**  
parsed games

**8,621**  
after  
preprocessing

**43**  
final features

### Preprocessing

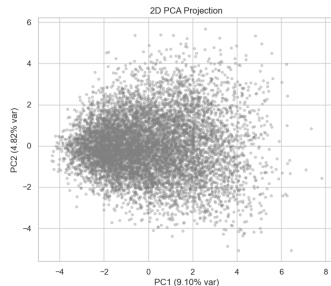
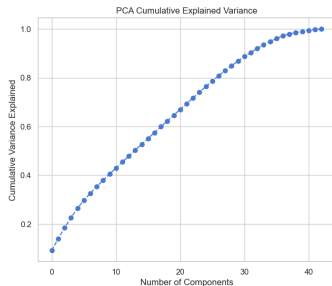
- Removed missing values.
- Converted ECO dummy columns to integers.
- Removed extreme rows using 1st and 99th percentiles.
- Applied  $\log_{1p}$  to skewed non-negative features.
- Standardized all numerical features.

### Important limitation

Outlier filtering removed **1,379 games**, about **13.79%** of parsed data.

- This improves distance-based clustering.
- But some real unusual tactical games may be removed.
- Results describe mainly typical games.

# PCA: First Look at Geometry



- PC1 explains **9.10%**.
- PC2 explains **4.82%**.
- Together: only **13.92%**.
- 2D projection looks like one broad cloud.

## Interpretation

Chess games do not form clean separated islands in first two principal components.

## Benchmark Clustering Results

**0.0935**

K-Means K=3

**0.0489**

GMM K=3

**0.0761**

Agglomerative K=3

**0.0776**

DBSCAN core clusters

### Main interpretation

All full-space clustering scores are low. This suggests weak global separation, not just failure of one algorithm.

*The clusters should be interpreted as tendencies, not exact labels like tactical or positional.*

| Group         | Features | Silhouette |
|---------------|----------|------------|
| Dynamic       | 7        | 0.2777     |
| ECO / Opening | 20       | 0.2487     |
| Structural    | 16       | 0.1106     |
| All Features  | 43       | 0.0935     |

### Key result

Dynamic features give strongest clustering signal.

- Captures, checks, and volatility separate games better.
- Opening family also creates strong structure.
- Structural features help, but are weaker.
- Mixing all features lowers global separation.

### Original expectation

Pawn structure and center control might produce clear chess styles.

- Structural features did have signal.
- But they did not separate games as strongly as dynamic features.

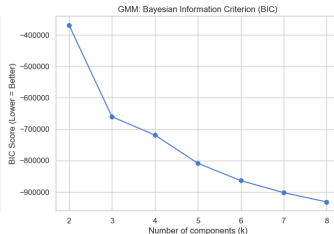
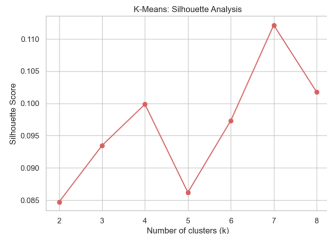
### Final interpretation

The full feature space mixes different kinds of chess information.

- Some dimensions create useful separation.
- Other dimensions add noise or overlap.
- Chess style behaves more like spectrum.

**This became the main research finding of the project.**

# K-Means and GMM Tuning

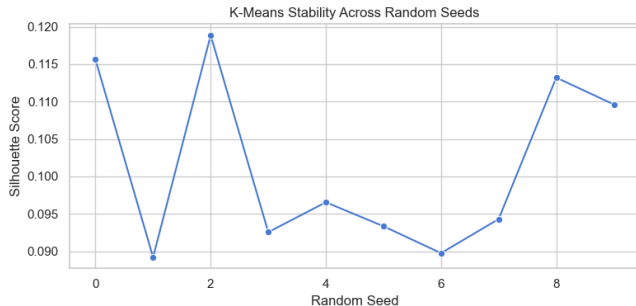


- K-Means silhouette was tested for  $K = 2$  to 8.
- Best K-Means value was **K = 7**.
- Best silhouette was still only around **0.11**.
- GMM BIC was also checked for comparison.

## Interpretation

$K = 7$  is practical modeling choice, not proof of seven natural chess styles.

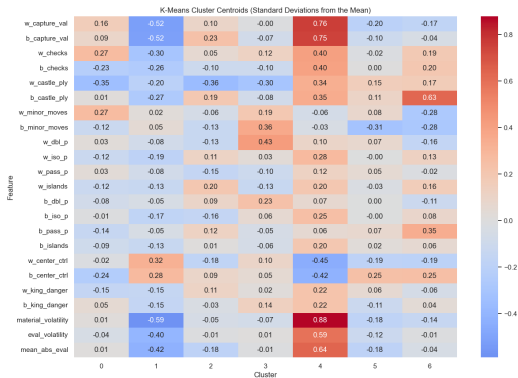
## K-Means Stability Across Seeds



### Stability result

- Mean silhouette: **0.1013**
- Std across seeds: **0.0116**
- Scores stay in same low range.
- Weak separation is not one unlucky initialization.
- K-Means clusters are not perfectly stable either.

# Cluster Profiling by Centroids



- Centroids are standardized deviations from mean.
- Positive values mean above-average feature level.
- Some clusters show high capture values and volatility.
- Other clusters look quieter and more stable.

## Main point

K-Means finds tendencies, not clean chess styles.

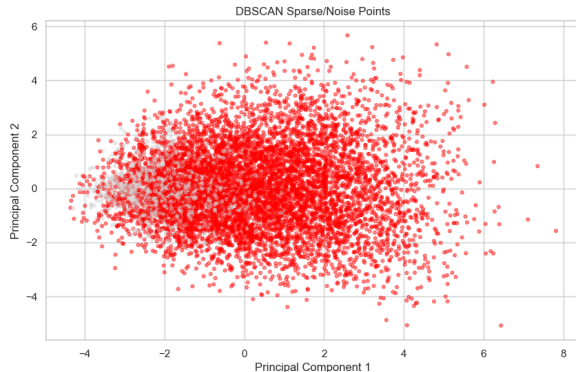
### Why DBSCAN?

- Does not force every point into cluster.
- Can identify sparse or unusual regions.
- Useful for checking density structure.

### Final setting

- `eps` = 2.8
- `min_samples` = 15
- 4 core clusters
- 84.35% sparse/noise points

*We do not interpret 84.35% as true anomalies. It means broad dense regions are hard to find.*



- Red points are sparse/noise points.
- Most games are outside dense DBSCAN core.
- Sparse group has higher capture values and volatility.
- This supports continuous feature-space interpretation.

## Interpretation

DBSCAN is not best general clustering model here, but it supports the main conclusion.

| Model         | Main Role               | Strength                            | Interpretation               |
|---------------|-------------------------|-------------------------------------|------------------------------|
| K-Means       | Main baseline           | Easy centroid interpretation        | Useful for broad tendencies  |
| GMM           | Soft clustering         | Probabilistic assignment            | Confirms overlap             |
| Agglomerative | Hierarchical comparison | No centroid initialization          | Similar weak separation      |
| DBSCAN        | Sparse-region detection | Finds dense cores and sparse points | Better for density structure |

### Overall conclusion

No method produced strongly separated clusters in full feature space.

| Component      | Main Result                | Interpretation               |
|----------------|----------------------------|------------------------------|
| Dataset        | 10,000 parsed, 8,621 final | Large enough local study     |
| Features       | 43 after ECO encoding      | Meets feature requirement    |
| PCA            | $PC1 + PC2 = 13.92\%$      | No clear 2D separation       |
| Benchmark      | K-Means = 0.0935           | Weak full-space clusters     |
| Feature groups | Dynamic = 0.2777           | Stronger subspace signal     |
| K tuning       | Best K = 7                 | Detail, not clean styles     |
| DBSCAN         | 84.35% sparse/noise        | Continuous density structure |

**Main finding:** chess styles behave like spectrum, not strict labels.

### Original plan

- First 15 moves.
- K-Means, DBSCAN, PCA.
- Engineered chess features.

### Final implementation

- First 20 full moves.
- 43 final features.
- Added GMM and Agglomerative.
- Added feature group comparison.
- Added stability and DBSCAN tuning.

*The main direction stayed same: unsupervised study of chess structures without outcome labels.*

- ① Full feature space has weak cluster separation.
- ② Dynamic features give strongest clustering signal.
- ③ ECO/opening family also creates strong structure.
- ④ Structural features are weaker but still useful.
- ⑤ DBSCAN is not good general clustering model here.
- ⑥ Chess styles behave like continuous spectrum.

### Strongest result

The project did not find perfect chess styles. It found that full feature space is messy, but dynamic activity and opening family reveal partial structure.

### Data and features

- Features are manually engineered.
- Long-term plans and coordination are hard to encode.
- Stockfish evaluations were sampled only at selected points.

### Interpretation

- Outlier filtering removes some unusual games.
- ECO is not same as chess style.
- No ground-truth labels for style.

- We built an unsupervised pipeline from 10,000 Lichess games.
- Final dataset used 8,621 games and 43 features.
- Full feature space produced weak clustering.
- Feature groups revealed stronger partial structure.
- Dynamic features and opening family were most informative.
- Chess styles are better understood as tendencies on spectrum.

**Thank you**